

Your grade: 80%

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1.

When using a Kalman filter for parameter estimation, the parameter-model "state" equation has a linear form, which simplifies the computations in several SPKF steps compared to the SPKF for a fully nonlinear model. Which step(s) are simplified? Select all that apply.

☐ Step 1a: Parameter prediction time update.

☐ Step 1b: Prediction error covariance time update.

☒ Step 1c: Predict system output.

⊗ This should not be selected

No. The output equation is still potentially nonlinear, and the need to generate sigma points as input to this equation makes this step more complex than before.

☐ Step 2a: Estimator gain matrix.
- 1 / 1 point

2.

Which parameter-estimation SPKF steps remain essentially unchanged in conceptual form when compared to the final equations that apply SPKF to estimate a model's state.

☐ Step 1c: Predict system output.

☒ Step 2a: Estimator gain matrix.

✔ Correct

Yes. This is unchanged in principle.

☒ Step 2b: Parameter estimate measurement update.

✔ Correct

Yes. This is unchanged in principle.

☒ Step 2c: Estimation error covariance measurement update.

✔ Correct

Yes. This is unchanged in principle.

1 / 1 point

3.

Execute the Coursera Lab Jupyter notebook for Lesson 3.4.2, which implements the SPKF for parameter estimation.

What is the RMS parameter-estimation error for the "k" parameter of the model? Enter your result with two digits to the right of the decimal point.

0.14

✔ Correct

Yes. That is correct.

1 / 1 point

4.

For the same Coursera Lab Jupyter notebook for Lesson 3.4.2, what is the percentage of time that the "k" parameter estimation error is outside of its estimation confidence bounds? Enter your answer with one digit to the right of the decimal point.

0.1

✔ Correct

Yes, that is correct.

1 / 1 point

5.

In a parameter-estimation filter, if the estimate $\hat{\theta}$ is not adapting quickly enough to track changes to the true value of θ , what do you think you should do?

☐ Decrease $\Sigma_{\hat{\theta}}$.

☐ Increase $\Sigma_{\hat{\theta}}$.

☐ Decrease $\Sigma_{\hat{r}}$.

☒ Increase $\Sigma_{\hat{r}}$.

✔ Correct

Yes. This will help the filter to adapt more quickly.

1 / 1 point

6.

In a parameter-estimation filter, if you have high confidence that your initial estimate for $\hat{\theta}_0$ is very good, how should you set your covariance matrices?

☐ You should have small values in $\Sigma_{\hat{\theta}}$.

☐ You should have large values in $\Sigma_{\hat{r}}$.

☒ You should have small values in $\Sigma_{\hat{\theta},0}^+$.

☐ You should have large values in $\Sigma_{\hat{\theta}}$.

☐ You should have large values in $\Sigma_{\hat{\theta},0}^+$.

☐ You should have small values in $\Sigma_{\hat{r}}$.

✔ Correct

Yes. The uncertainty of $\hat{\theta}$ is small, so we should have small values.

1 / 1 point

7.

Execute the Coursera Lab Jupyter notebook for Lesson 3.4.4. What is the RMS parameter-estimation error for the "m" parameter of the model? Enter your result with two digits to the right of the decimal point.

0.85

✔ Correct

Yes. That is correct.

1 / 1 point

8.

For the same Coursera Lab Jupyter notebook for Lesson 3.4.4, what is the percentage of time that the "m" parameter estimation error is outside of its estimation confidence bounds? Enter your answer with two digits to the right of the decimal point.

0.11

✔ Correct

Yes, that is correct.

1 / 1 point

9.

When implementing dual estimation using EKF, which total derivative(s) must be computed recursively?

☒ Only $dx_k/d\theta$, as with parameter estimation.

☐ Only $\hat{C}_k^\theta = dh(\cdot)/d\theta$.

☐ Both $dx_k/d\theta$ and $\hat{C}_k^\theta = dh(\cdot)/d\theta$.

☐ We must compute $dh(\cdot)/d\theta$, $d\hat{x}_k^-/d\theta$, and $d\hat{x}_{k-1}^+/d\theta$.

⊗ Incorrect

No. In dual estimation, we don't know x_k , only a prediction or estimate thereof.

1 point

10.

Execute the Coursera Lab Jupyter notebook for Lesson 3.4.6, which implements both EKF and SPKF for joint state and parameter estimation.

Which filter gives lower root-mean-squared error on the position-state estimate?

☐ The joint EKF.

☒ The joint SPKF.

☐ Neither. Both filters give identical RMS error.

✔ Correct

Yes, that is correct.

1 / 1 point